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Rock Paper Scissors Game Report

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1 Contents

1.	Introduction	3
2.	Project Objectives.....	3
3.	Technologies Used	4
4.	Game Description.....	4
5.	Project Features	5
6.	System Implementation.....	5
7.	Example Code Snippets.....	6
8.	Challenges Faced.....	6
9.	Future Improvements	7
10.	Conclusion.....	7

1. Introduction

This project is a desktop application developed using the Python programming language and the Tkinter graphical user interface library. The application is based on the classic Rock Paper Scissors game between the player and the computer.

The project was designed to support both Arabic and English languages within the same interface while providing a modern and interactive user experience. The application includes several features such as a countdown timer, score tracking system, draw counter, automatic loss when time expires, and an executable desktop version (.exe).

The primary objective of this project is not merely to create a simple entertainment game, but rather to enhance programming knowledge and software development skills through practical implementation and experimentation. The project focuses on applying programming concepts in a real-world environment, including graphical user interface development, event handling, timer management, and application deployment.

2. Project Objectives

The objectives of this project include:

- ❖ Learning and practicing Python programming
- ❖ Developing graphical user interfaces using Tkinter
- ❖ Understanding event-driven programming concepts
- ❖ Implementing timers and score systems
- ❖ Designing bilingual interfaces
- ❖ Converting Python applications into executable desktop programs
- ❖ Improving debugging and problem-solving skills
- ❖ Gaining practical experience in software implementation

3. Technologies Used

The following technologies and tools were used during the development process:

- ❖ Python Programming Language
- ❖ Tkinter GUI Library
- ❖ Random Library
- ❖ PyInstaller
- ❖ Windows Command Prompt (CMD)

4. Game Description

The game allows the player to choose one of the following options:

- ❖ Rock
- ❖ Paper
- ❖ Scissors

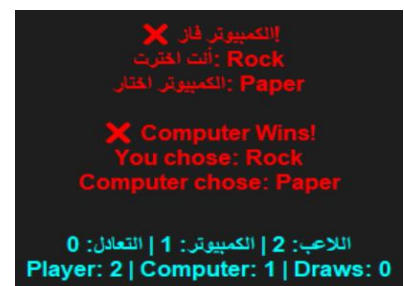
The computer then randomly selects another option. The application compares both choices and determines the result of the round:

- ❖ Win
- ❖ Lose
- ❖ Draw

The application also contains a countdown timer. If the player does not make a selection before the timer reaches zero, the player automatically loses the round and the computer receives one point.

The interface displays:

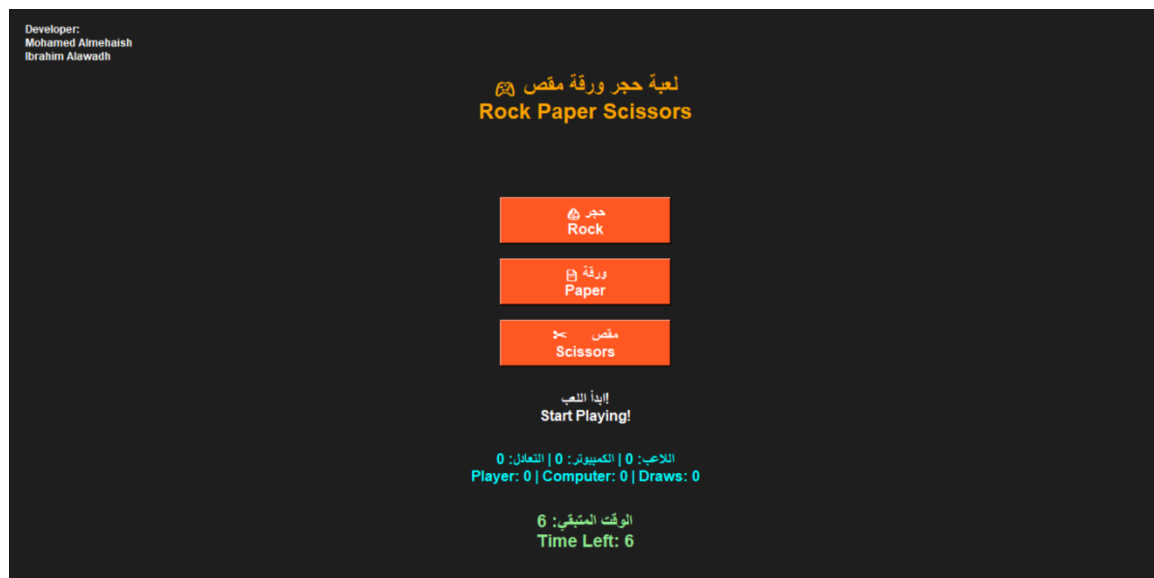
- ❖ Current game result
- ❖ Remaining time
- ❖ Player score
- ❖ Computer score
- ❖ Draw count



5. Project Features

The project includes several features, including:

- ❖ Modern graphical user interface
- ❖ Arabic and English language support
- ❖ Countdown timer system
- ❖ Automatic score calculation
- ❖ Draw counter
- ❖ Interactive buttons and controls
- ❖ Executable desktop application (.exe)
- ❖ Responsive and user-friendly interface



6. System Implementation

The application was implemented using Python and Tkinter. Different programming concepts were applied during development, including:

- ❖ Functions
- ❖ Conditional statements
- ❖ Loops
- ❖ Variables
- ❖ Event handling
- ❖ GUI design

The game logic depends on comparing the player's choice with the computer's randomly generated selection.

7. Example Code Snippets

1. Random Computer Choice: `computer_choice = random.choice(choices)`

This line allows the computer to randomly select one of the available game options.

2. Timer Function :

```
if time_left > 0:  
    time_left -= 1  
    window.after(1000, update_timer)
```

```
if time_left > 0:  
    time_left -= 1  
    window.after(1000, update_timer)
```

This code decreases the timer every second and continuously updates the interface.

3. Score System: `player_score += 1`

This statement increases the player score whenever the player wins a round.

8. Challenges Faced

Several challenges were encountered during the development process, including:

- ❖ Managing Arabic text within the graphical interface
- ❖ Implementing countdown timer functionality
- ❖ Dynamically updating scores and game states
- ❖ Designing a bilingual graphical interface
- ❖ Converting the Python project into an executable desktop application
- ❖ Handling command-line operations using CMD

9. Future Improvements

Possible future enhancements for the project include:

- ❖ Adding sound effects and background music
- ❖ Developing multiplayer mode
- ❖ Creating an online web-based version
- ❖ Adding user authentication and player profiles
- ❖ Saving player statistics and game history
- ❖ Improving interface animations and visual effects

10. Conclusion

This project provided valuable practical experience in Python programming and graphical user interface development. It contributed to improving programming logic, debugging abilities, problem-solving skills, and software packaging knowledge.

The project represents a practical educational experience rather than simply creating a small game. It demonstrates the importance of learning through implementation, experimentation, and continuous development in software engineering and application design.